QUANT

Mensuration & Geometry

(Circles, Quadrilaterals & Trigonometry)

Assignment 01

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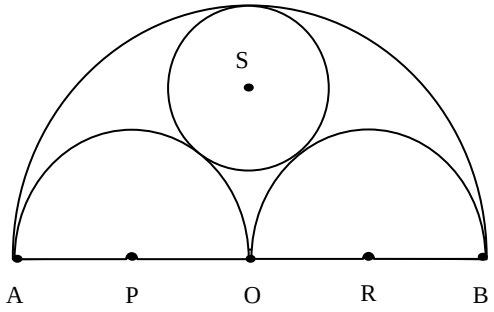
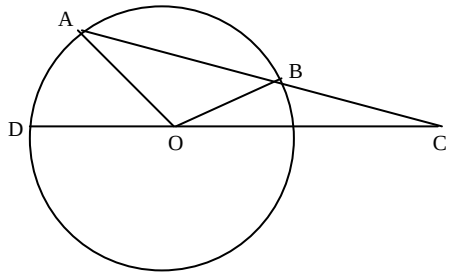
Geometry Online Assignment Cluster II -01

1. Three horses are grazing within a semi-circular field. In the diagram given below, AB is the diameter of the semi-circular field with centre at O. Horses are tied up at P, R and S such that PO and RO are the radii of semicircles with centres at P and R respectively, and S is the centre of the circle touching the two semi-circles with diameters AO and OB. The horses tied at P and R can graze within the respective semi-circles and the horse tied at S can graze within the circle centered at S. The percentage of the area of the semi-circle with diameter AB that cannot be grazed by the horses is nearest to



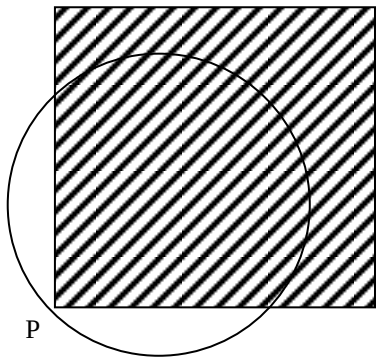
1. 20 2. 28 3. 36 4. 40

2. In the figure given below, AB is the chord of a circle with centre O. AB is extended to C such that BC = OB. The straight line CO is produced to meet the circle at D. If $\angle ACD = y$ degrees and $\angle AOD = x$ degrees such that $x = ky$, then the value of k is



1. 3 2. 2 3. 1 4. None of these

DIRECTIONS *for Questions 3 & 4:* A punching machine is used to punch a circular hole of diameter two units from a square sheet of Aluminium of width 2 units, as shown below. The hole is punched such that the circular hole touches one corner P of the square sheet and the diameter of the hole originating at P is in line with a diagonal of the square :



3. The proportion of the sheet area that remains after punching is

1. $(\pi+2)/8$ 2. $(6-\pi)/8$ 3. $(4-\pi)/4$ 4. $(\pi-2)/4$

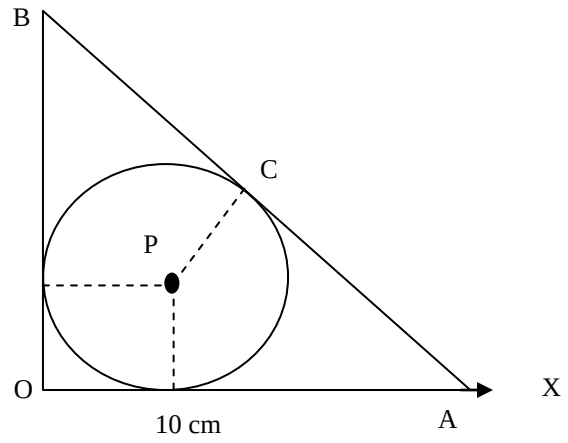
4. Find the area of the part of the circle (round punch) falling outside the square sheet.

1. $\pi/4$ 2. $(\pi-1)/2$ 3. $(\pi-1)/4$ 4. $(\pi-2)/2$

5. There is a circle of radius 1 cm. Each member of a sequence of regular polygons S1(n), n = 4, 5, 6, ... where n = number of sides of the polygon, is circumscribing the circle and each member of the sequence of regular polygons S2(n), n = 4, 5, 6, ..., where n is the number of sides of the polygon, is inscribed in the circle. Let L1(n) and L2(n) denote perimeters of the corresponding polygons of S1(n) and S2(n). Then $\frac{[L1(13)+2\pi]}{L2(17)}$ is :

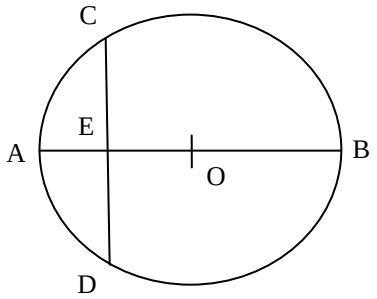
1. Greater than $\pi/4$ but less than 1
2. Greater than 1 and less than 2
3. Greater than 2
4. Less then $\pi/4$

6. A circle of radius 3 cm is drawn touching both the axes OX and OY as shown is the figure. A tangent is drawn to the circle from a point A, 10cm away from O on OX, which meets OY at B. find the length of OB.



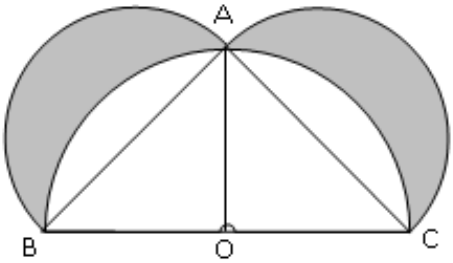
1. 10 cm 2. $\frac{20}{7}\sqrt{58}$ cm 3. $\frac{21}{2}$ cm 4. None of these

7. In the circle shown below, AB is a diameter and CD is a chord. The lengths of AB and CD are integers, such that the length of CD is obtained by reversing the two digits of length of AB. If the length of OE is a rational number, find the length of OE.



1. 16 2. 16.5 3. 33 4. None of these

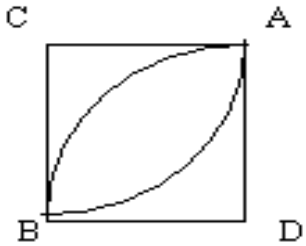
8.



In Δ ABC, a semicircle with centre O is drawn on side BC as shown. Similarly, semicircles are drawn on sides AB and AC. If BC = 14 cm, what is the area of the shaded region?

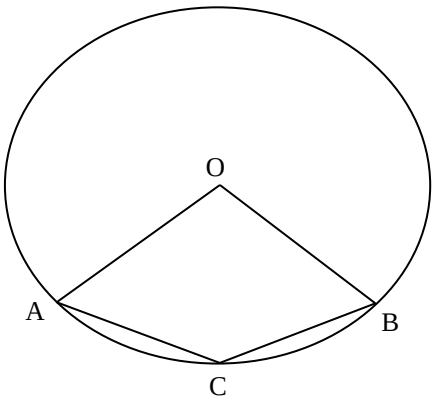
1. 22 cm^2 2. 259 cm^2 3. 49 cm^2 4. 154 cm^2

9. In a square of side 4 cm, two quadrants of a circle are drawn with centers on the vertices C and D and radius equal to the side of the square. Then the area of the petal AB is



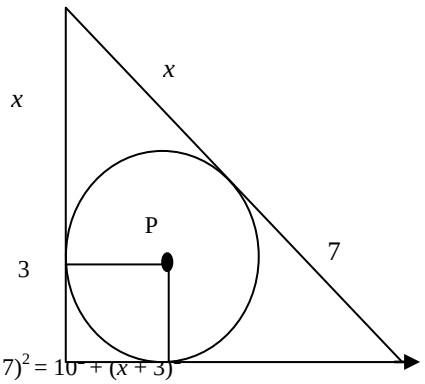
1. 12 cm^2 2. 7.76 cm^2 3. 9.12 cm^2 4. 8.93 cm^2

10. O is the centre of the circle. If angle ACB = 130 deg, find smaller angle AOB



1. 50^0 2. 100^0 3. 130^0 4. 210^0

Answer key & Explanations

Q. No.	Ans.	Exp.
1.	2	Let the radius of the medium circle be R and x be the radius of the smallest circle. Now $(2R - X)^2 + R^2 = (R + X)^2 \Rightarrow 4R^2 + X^2 - 4RX + R^2 = R^2 + X^2 + 2RX$. So $4R^2 = 6RX \Rightarrow R/X = 3/2$. Hence X = 2R/3. Area of big semi-circle = $½ \times \pi \times (2R)^2 = 2\pi R^2$. Area of two medium semi circles is πR^2. Area of small circle = $\pi \times (2R/3)^2 = 4\pi R^2/9$. Cancel πr^2 in all the three areas & get % area not grazed = 2 – 1 – 4/9 = (5/9 ÷ 2) × 100 = roughly 28%.
2.	1	Consider Δ BOC: ∠BCO = ∠COB = y°. Now ∠ABO = exterior ∠ of triangle BCO = y + y = 2y°. Here ∠BOA = 180 – (∠AOD + ∠COB). Consider Δ AOB; 180 – (x + y) + 2y + 2y = 180°, x + y = 4y Or x = 3y; so k = 3.
3.	2	Area after Punching = Area of Square ABCD – Area of Circle $(2\times 2 = 4) - (\pi \times 1^2 = \pi)$ + Area of the part outside Square Sheet Which is $\frac{\pi - 2}{2}$ (See the Calculation of this from the next Question) $= 4 - \pi + \frac{\pi - 2}{2} = \frac{8 - 2\pi + \pi - 2}{2} = \frac{6 - \pi}{2}$
4.	4	Required area = Area of Circle – Area of Square AEFG = $\frac{\pi \times 1 - \sqrt{2} \times \sqrt{2}}{2} = \frac{\pi - 2}{2}$
5.	3	Seemingly a complicated question. It is not. Can you write the expression as L1 (13)/L2 (17) + 2π/L2 (17) {Breaking the given expression into two parts}. Now L2 represents the perimeter of a 17-sided polygon inside the circle. Now the perimeter of any inscribed polygon will be less than circle, in which it is inscribed and obviously lesser than the polygon, in which that circle is inscribed. Hence both of the broken expressions are more than 1 and hence the answer is more than 2.
6.	3	 $(x + 7)^2 = 10^2 + (x + 3)^2$ (Tangents from the external point are equal) $\Rightarrow x = 7.5$cms $\Rightarrow$ OB = $x + 3 = 10.5$ cm
7.	2	Let AB = $10x + y \Rightarrow$ CD = $10y + x$ $\Rightarrow OE^2 = OC^2 - CE^2$ $= \left(\frac{1}{2}AB\right)^2 - \left(\frac{1}{2}CD\right)^2$ $\Rightarrow OE^2 = \left(\frac{10x + y}{2}\right)^2 - \left(\frac{10y + x}{2}\right)^2$ $\Rightarrow OE^2 = \frac{1}{2}\sqrt{99x^2 - 99y^2}$ $\Rightarrow OE^2 = \frac{3}{2}\sqrt{11(x + y)(x - y)}$

		Since length of OE is a rational number $\Rightarrow \sqrt{11(x^2 - y^2)}$ must be a perfect square. The only combination is $x = 6$, $y = 5$ (Since $x + y$ is the 0 to 9) $\therefore$ OE = $\frac{3}{2} \times 11 = 16.5$.
8.	3	The area of the entire figure is made up of the areas of the semicircles on sides AB and AC and the area of ΔABC. Since O is the centre, OB = OC = OA = radius = 7 cm. So, AB = AC = 7√2 cm. The sum of areas of the two semicircles on AB and AC is $\pi(7/\sqrt{2})^2 = 77$ cm^2. The area of ΔABC = ½ × 14 × 7 = 49 cm^2. So the area of the figure is 77 + 49 = 126 cm^2. Now, the area of the shaded region can be calculated as the area of the figure less the area of the semicircle on side BC. The area of the semicircle on side BC $= (\pi/2) \times 7^2 = 77$. Thus the area of the shaded region is 126 – 77 = 49 cm^2.
9.	3	Since the petal is formed by the overlap of the two quadrants, Area of the petal = area of the two quadrants – area of the square. Hence $2\frac{3.14 \times 4^2}{4} - 4^2$ = 9.12 sq. cm.
10.	2	Bigger arc AB = 260° (twice the angle made at the circumference). The remaining portion at the center is ∠AOB = 360 –260 =100°. Hence 2nd option.